

SCD Dbase Sorter - Project Documentation

Document: TECHNICAL_MANUAL.md

SCD Dbase Sorter - Technical Manual

1. Project Overview

2. Core Features

3. System Architecture

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4. Chronological Approval Log

| Date | Task ID | Member | Role | Status | Message Body |

Detail

| :--- |

5. Deployment Instructions

1. Ins

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6. Source Code Appendix

6.1 /home/team/shared/SCD_Dbase_Sorter/app.py

```
import streamlit as st
```

Add processor directory to path

Import team modules

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??

Page

??

Ses

??

Sid

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SMTP Configuration

if st.sidebar.button("Save SMTP Settings"):

st.sidebar.markdown("---")

Data Paths Info

Hospital Config

st.sidebar.markdown("---")

Security / PII Section

st.side

Documentation Delivery Section

st.side

from mailer import _send_email

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success

st.sidebar.markdown("----")

st.sidebar

??

Main

===== STEP 1: File Upload =====

st.header

File Password (for encrypted files) ? only show if checkbox is selected

upload

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if uploaded_file is not None:

st.ses

Preview the uploaded file

col1, c

Show first 5 rows

Har

st.dataframe(preview_df, use_container_width=True)

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with col2:

===== STEP 2: Process & Sort =====

if st.button("? Process & Sort", type="primary", use_container_width=True):

Step B: Process (append to master, generate hospital sheets)

Store in session state

st.success(f"? Processing complete! {len(mapped_df)} records processed.")

except Exception as e:

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else:

===== STEP 3: Data Overview & Visualization =====

if st.session_state.processing_complete and st.session_state.master_df is not None:

Use tabs for different views

with tab_overview:

st.markdown("---")

Records per Hospital chart

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if "Hos

Bar chart using st.bar_chart (built-in, no extra deps)

chart_

Show numeric table

col1, c

Records per Year chart

if "Yea

```
chart_year = year_counts.set_index("Year")
```

st.bar

```
with col2:
```

st.dat

Cross-tab: Hospital x Year

if "Hos

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Show PII security note

with tab_hospitals:

if selected_hospital:

Show validation status breakdown

with tab_raw:

elif os.path.exists(MASTER_DB_PATH):

if "Pat

st.sub

hosp_

if "Val

st.sub

Try

===== STEP 4: Email Actions =====

if st.session_state.processing_complete and st.session_state.master_df is not None:

st.hea

maste

if "Hospital" in master_df.columns:

hospi

Option to email all hospitals at once

col_b

Export hospital data to temp file for attachment

safe_

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```
result = send_validation_request(hospital, hosp_data, attachments=[temp_attach])
```

```
st.write("***Validation Request Results:**")
```

```
with col_bulk2:
```

```
safe_name = str(hospital).strip().replace(" ", "_")
```

```
result = send_finalized_data(hospital, hosp_data, attachments=[temp_attach])
```

```
st.write("***Finalized Data Send Results:**")
```

```
st.markdown("----")
```

Individual hospital email controls

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col_h

col_info1, col_info2 = st.columns(2)

with c

if st.button("? Send Now", type="primary"):

if sele

if email_type == "Validation Request":

succe

if success:

st.suc

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else:

st.info

Show existing database data for emailing, even if no new data was processed

if os.p

if st.button("? Send Validation Request"):

hosp

??

Foc

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6.2 /home/team/shared/SCD_Dbase_Sorter/processor/mapping.py

```
from sanitization import sanitize_dataframe
```

Standard Master Headings

Alias Mapping

```
def _get_excel_data(file_path, password=None):
```

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```
def find_master_match(header_text):
```

```
"""Che
```

```
clean_text = str(header_text).strip().lower()
```

Check exact/case-insensitive master headings

```
for ma
```

Check aliases

```
for ma
```

```
return None
```

```
def get_column_mapping(file_path, password=None):
```

```
"""
```

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```
mapping = {}
```

```
for col in range(num_cols):
```

Try Row 1

Try Row 2

```
return mapping
```

```
CONFIG_PATH = "/home/team/shared/SCD_Dbase_Sorter/data/config/hospital_emails.csv"
```

```
def load_hospital_config():
```

```
def mask_pii_data(df):
```

```
def _mask_string(val):
```

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if not

```
def load_and_map_data(file_path, password=None):
```

.....

```
excel_data = _get_excel_data(file_path, password)
```

Rea

Rename columns based on mapping

df_ma

```
for col_idx, master_name in mapping.items():
```

if col_

Add default values for missing master columns

for ma

Apply PII masking

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Apply Sanitization

Set default Validation_Status if missing

Set Date_Added

```
audit_logger.log_action("LOAD_AND_MAP", details={"file": file_path, "records": len(df_mapped)})
```

Fill Validator_Email and Hospital_Email from config if missing

Fill missing Validator_Email

Fill missing Hospital_Email

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return df_mapped

'''

6.3 /home/team/shared/SCD_Dbase_Sorter/processor/sorter.py

MASTER_DB_PATH = "/home/team/shared/SCD_Dbase_Sorter/data/master/Master_Database.xlsx"

def ensure_directories():

def update_master_database(new_data_df):

if os.path.exists(MASTER_DB_PATH):

Ensure Date_Added is datetime

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```
combined_df = pd.concat([master_df, new_data_df], ignore_index=True)
```

except

Save and Encrypt

combined

```
audit_logger.log_action("UPDATE_MASTER", details={"records_added": len(new_data_df)})
```

return

```
def generate_hospital_sheets(master_df):
```

.....

```
if master_df.empty:
```

return

Group by Hospital

group

Decrypt Master for modification

decry

We'll use ExcelWriter to update the Master_Database with sheets

If fi

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with pd.ExcelWriter(MASTER_DB_PATH if not master_buffer else master_buffer, **writer_kwargs) as writer:

if mas

Simplified approach: Overwrite everything since we have the full master_df

with p

for hospital, group in grouped:

if pd.is

Create a safe filename and sheet name

safe_

Sort by Year

if 'Yea

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1. Save to individual file

file_pa

2. Save to a sheet in Master_Database

sheet,

```
print(f"Generated/Updated sheet and file for {hospital_name}")
```

Finally encrypt the Master Database

encrypt

```
def process_new_data(new_data_df):
```

.....

2. Refresh hospital files

gener

```
return master_df
```

```
...
```

6.4 /home/team/shared/SCD_Dbase_Sorter/processor/mailer.py

```
```pyth
```

# SCD Dbase Sorter - Project Documentation

```
import smtplib
```

```
import
```

**Configuration file path**

```
HOSP
```

**SMTP Configuration (dummy defaults, can be overridden)**

```
SMTP
```

```
def load_hospital_emails():
```

```
.....
```

## SCD Dbase Sorter - Project Documentation

```
df = pd.read_csv(HOSPITAL_EMAILS_PATH)
```

```
return
```

```
def get_hospital_email(hospital_name):
```

```
"""Ret
```

```
def get_validator_email(hospital_name):
```

```
"""Ret
```

```
def get_all_hospitals():
```

```
"""Ret
```

```
def get_hospital_data_path(hospital_name):
```

```
"""
```

# SCD Dbase Sorter - Project Documentation

def load\_hospital\_data(hospital\_name):

"""

Args:

hospi

Returns:

DataF

def \_create\_smtp\_connection():

"""Cre

def \_send\_email(to\_email, subject, body, attachments=None):

"""

Args:

to\_em

# SCD Dbase Sorter - Project Documentation

Returns:

bool:

```
msg = MIMEMultipart()
```

msg[''

**Attach body as plain text**

msg.a

**Attach files if provided**

if atta

```
part = MIMEBase('application', 'octet-stream')
```

part.s

try:

serve

# SCD Dbase Sorter - Project Documentation

def send\_validation\_request(hospital\_name, hospital\_data\_df=None, attachments=None):

.....

Args:

hospit

Returns:

bool: "

## Auto-attach hospital's Excel file if no attachments provided

if atta

subject = f"Validation Request - {hospital\_name} SCD Data"

# SCD Dbase Sorter - Project Documentation

A new dataset has been submitted for the following hospital and requires validation:

Hospital: {hospital\_name}

Please review the attached data and confirm its accuracy by replying to this email.

If you have any questions, please contact the database administrator.

Best regards,

```
def send_finalized_data(hospital_name, hospital_data_df=None, attachments=None):
```

Args:

Returns:

**Auto-attach hospital's Excel file if no attachments provided**

# SCD Dbase Sorter - Project Documentation

```
subject = f"Finalized SCD Data - {hospital_name}"
```

Please find attached the finalized SCD (Sickle Cell Disease) data for your review and records.

Hospital: {hospital\_name}

If you have any questions or require further assistance, please do not hesitate to reach out.

Best regards,

```
def send_bulk_validation_requests(hospital_data_dict):
```

Args:

Returns:



# SCD Dbase Sorter - Project Documentation

def send\_bulk\_finalized\_data(hospital\_data\_dict):

.....

Args:

hospit

Returns:

dict: S

def configure\_smtp(host=None, port=None, username=None, password=None, from\_email=None, from\_name=None):

.....

Args:

host:

# SCD Dbase Sorter - Project Documentation

=====

===

def export\_df\_to\_excel(df, output\_path):

if \_\_name\_\_ == "\_\_main\_\_":

## 6.5 /home/team/shared/SCD\_Dbase\_Sorter/processor/encryption.py

KEY\_FILE = "/home/team/shared/SCD\_Dbase\_Sorter/data/config/.master.key"

def ensure\_key():

# SCD Dbase Sorter - Project Documentation

```
def encrypt_file(file_path):
```

```
"""Enc
```

```
 key = ensure_key()
```

```
 fernet
```

```
 with open(file_path, "rb") as f:
```

```
 data =
```

```
 encrypted_data = fernet.encrypt(data)
```

```
 with open(file_path, "wb") as f:
```

```
 f.write
```

```
def decrypt_file_to_memory(file_path):
```

```
"""Dec
```

```
 key = ensure_key()
```

```
 fernet
```

```
 with open(file_path, "rb") as f:
```

```
 encry
```

```
 try:
```

# SCD Dbase Sorter - Project Documentation

decry

...

## 6.6 /home/team/shared/SCD\_Dbase\_Sorter/processor/logger.py

```pyth

LOG_DIR = "/home/team/shared/SCD_Dbase_Sorter/data/logs"

AUDI

class AuditLogger:

def __

Avoid adding multiple handlers if already initialized

if not

def log_action(self, action, user="system", details=None):

.....

Global instance

'''

6.7 /home/team/shared/SCD_Dbase_Sorter/processor/sanitization.py

def sanitize_value(val):

1. Prevent Excel Formula Injection

2. Prevent XSS by escaping HTML tags

return val

def sanitize_dataframe(df):

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...

6.12 /home/team/shared/SCD_Dbase_Sorter/processor/pdf_generator.py

```
class SCD_PDF(FPDF):
```

```
def footer(self):
```

```
def convert_md_to_pdf(md_path, pdf_path):
```

```
'''pyth
```

```
def he
```

```
self.se
```

```
.....
```

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```
pdf = SCD_PDF()
```

```
filename = os.path.basename(md_path)
```

```
effective_page_width = pdf.w - 2 * pdf.l_margin
```

```
with open(md_path, 'r', encoding='utf-8') as f:
```

```
if not line:
```

Basic Header handling

```
pdf.output(pdf_path)
```

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6.13 /home/team/shared/SCD_Dbase_Sorter/processor/search_bot.py

```
```pyth
```

```
class SearchBot:
```

```
... (l
```